

MURALI SAI BUDDAKKAGARI VENKATA

AI ENGINEER

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EDUCATION

Stevens Institute of Technology

Master of Science, Applied Artificial Intelligence

May 2026

NJ, US

PES University

Bachelor of Technology, Computer Science (Specialization in ML)

May 2024

KA, IND

TECHNICAL SKILLS

Programming Languages: Python, SQL

AI & Machine Learning: LLM Application Development, **Agentic Systems & Tool Orchestration (MCP)**, **Retrieval-Augmented Generation** (Chunking, Reranking, Grounding), **Prompt Engineering** (Chain-of-Thought, Few-Shot), **Function Calling**, **Multi-Agent Systems**, **Agent Evaluation (RAGAS)**, Guardrails & Reliability, **Multimodal AI**, **NLP**, **Embeddings**, **Vector Databases**, **LangChain**, **LangGraph**, **PyTorch**, **TensorFlow**, **HuggingFace**

Infrastructure & DevOps: **FastAPI**, **ChromaDB**, **Docker**, **OpenTelemetry** (Observability), **Google Cloud Platform**, **Azure**, **PostgreSQL**, **GitHub Actions**, **CI/CD**, **Git**, **REST APIs**, **ReactJS**, **OpenAI API**, **Claude API**, **Gemini API**

EXPERIENCE

Querkey Inc

Data Science Intern

Jun 2025 - Aug 2025

CA, US

- Developed and deployed a proof-of-concept real-time Patient Flow Tracking System using **multimodal AI** (Gemini 2.5 Flash) with **chain-of-thought** and **few-shot prompting** to improve routing efficiency, integrating a **RAG** pipeline that structured unstructured patient notes for real-time clinical insight retrieval.
- Cut **production ML** workflow failures 60% across 100K+ records by deploying a **Dockerized** observability stack (**MLflow**, **FastAPI**, **ChromaDB** vector store, **HuggingFace** monitoring), enabling real-time failure detection.
- Authored a 30-query **SQL** bank against the hospital's data to ground an **AI** chatbot's **RAG**-based answers to natural-language questions, and built a **Superset REST API** integration that generates and renders dashboard charts on demand from parameterized JSON payloads.

VIEW GROUP INDIA (FORMERLY VIEW SYNERGY)

AI and Automation Intern

Jan 2024 - Apr 2024

KA, IND

- As part of a **cross-functional** project team modernizing a live **production** platform, reviewed **React** frontend, **.NET Azure** Functions **API**, and **MySQL** stored-procedure code, tracing 2 recurring exceptions to root cause and surfacing **SQL**-injection vulnerabilities that fed into a 94-issue security assessment.
- Built and tested a new Roles/Designations **CRUD** feature end to end, including new **MySQL** stored procedures, a **.NET API** layer, and a **React** frontend, while documenting Microsoft Entra ID role setup for a proposed identity migration.
- Audited **Redis** usage across all 6 backend microservices and documented **Microsoft Entra ID** app registration and role setup, supporting a proposal to replace a fully custom **JWT/BCrypt** authentication system.

PROJECTS

SEC EDGAR RAG Filing Analyzer

- Deployed a production **RAG** system chunking and indexing 8,232 **SEC 10-K** filing chunks across 5 companies, verified by a 504-test regression suite run in **CI** and a 54-question **eval** suite (0.853 faithfulness, 0.936 citation accuracy, 0.981 refusal correctness), by building a **FastAPI** + **ChromaDB embeddings** pipeline with **cross-encoder reranking**, **RBAC**-enforced information barriers (FINRA Rule 2241 / SEC Rule 15g-1), and a serverless **Google Cloud Run** deployment (scale-to-zero), solo-built using **AI-assisted engineering** practices (**Claude Code**).
- Built literal **entity detection** pinning retrieval to the named company's filing at the database level, per-entity retrieval budgets for comparative questions, and a bounded **agentic** retry with **prompt-injection** and MNPI/ investment-advice output **guardrails**.

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Shipped a production **multi-agent** system (19 **REST** endpoints) that reasons, plans, and executes multi-step tasks end-to-end, validated by 106/113 passing tests against a live deployment, by architecting a 3-layer **LangGraph** hierarchy (planning supervisor, parallel specialists, reflective reviewer) with multi-model routing (GPT-4o, Claude Haiku) and two-tier **Redis** + **Supabase pgvector** memory.
- Built a custom **MCP** (Model Context Protocol) server integrating internal and third-party tools via **APIs** across 6 tools, instrumented end to end with **OpenTelemetry** distributed tracing, and a **Human-in-the-Loop** system spanning 4 escalation severity levels with full audit-trail logging, independently architected and shipped end-to-end using **spec-driven development** practices in **Claude Code**.

PUBLICATIONS AND CERTIFICATIONS

- Adaptive Ensemble Detection for Medical Imaging, M.S. Capstone, Stevens Institute of Technology (Advised by Prof. Kevin Lu). A size-conditioned Weighted Box Fusion ensemble of Faster R-CNN, Mask R-CNN, and YOLO26 for cell-nucleus detection, reaching 0.71 mAP@0.5 on the 2018 Data Science Bowl benchmark.
- Best Research Paper Award, ICICC 2024 - WorkFit: Real-time Multimodal Posture Monitoring System using CNN-based Computer Vision and Flex Sensor Fusion.